Cross-mission expiMap analysis recovers established tissue responses and identifies complementary pathway shifts in mouse spaceflight transcriptomes

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ABSTRACT

Spaceflight affects multiple organs, but mission differences can obscure responses that recur across studies. We mapped NASA OSDR mouse bulk RNA-seq samples into tissue-matched expiMap references trained on ARCHS4 and constrained by current mouse Reactome programs. Project-balanced flight-ground shifts were checked with conventional enrichment, held-out projects, three complete trainings, composition proxies, and member-gene review. Thymus, skin, liver, and spleen produced reproducible tissue-specific patterns; spleen had the strongest multi-pathway evidence.

INTRODUCTION

PROJECT OBJECTIVE

Learn how spaceflight changes living systems by asking which gene programs shift consistently across missions.

THE CENTRAL CHALLENGE

Observed expression = Spaceflight biology + Tissue biology + Animal variation + Study / protocol

Study identity and protocol can dominate an unconstrained expression representation. Tissue-matched reference mapping, accession conditioning, and equal project weighting reduce this bias without claiming to remove all mission confounding.

FINAL ANALYSIS SCOPE

Tissue	ARCHS4	OSDR	Projects	Programs
Thymus	1,362	117	5	387
Skin	2,593	151	4	319
Liver	5,000	197	9	364
Spleen	6,289	100	5	360

OSDR counts are primary analysis samples. Spleen excludes one condition-strain-confounded project.

ROBUSTNESS GATE

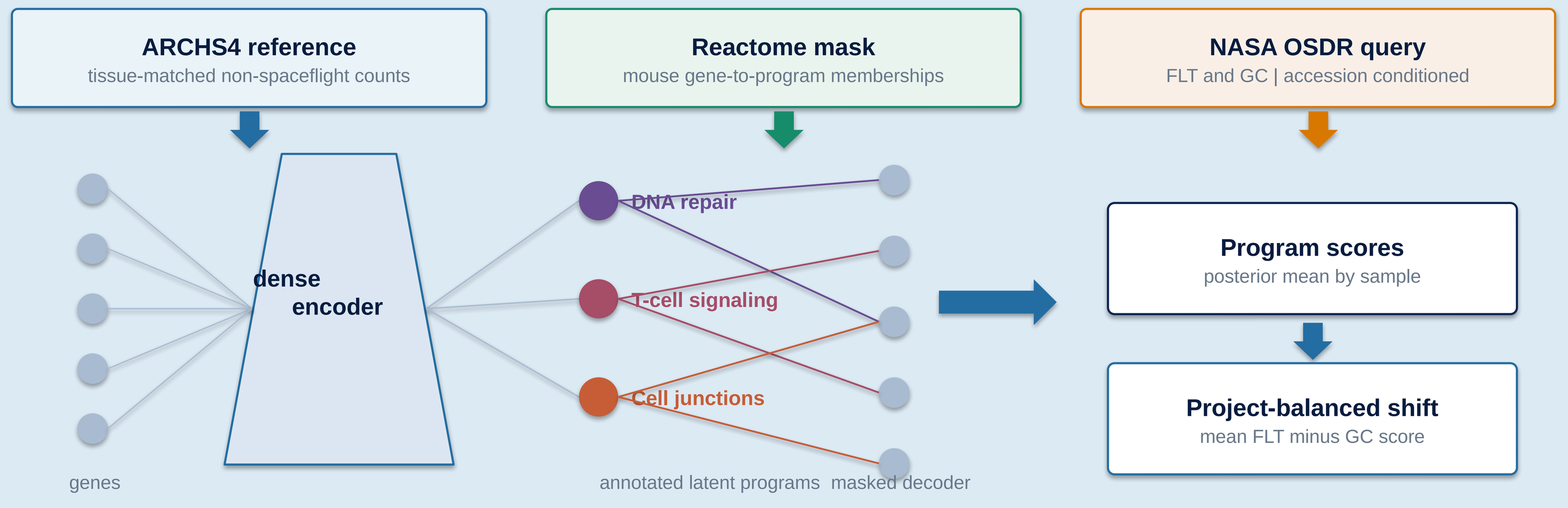
- ssGSEA + GSEA
- Held-out projects
- 3 full trainings
- Composition proxies
- Member genes
- Literature review

Complementary means an annotated Reactome program that adds a plausible perspective beyond the dominant phenotype in prior literature. De novo nodes were not retained in the final models.

METHODS

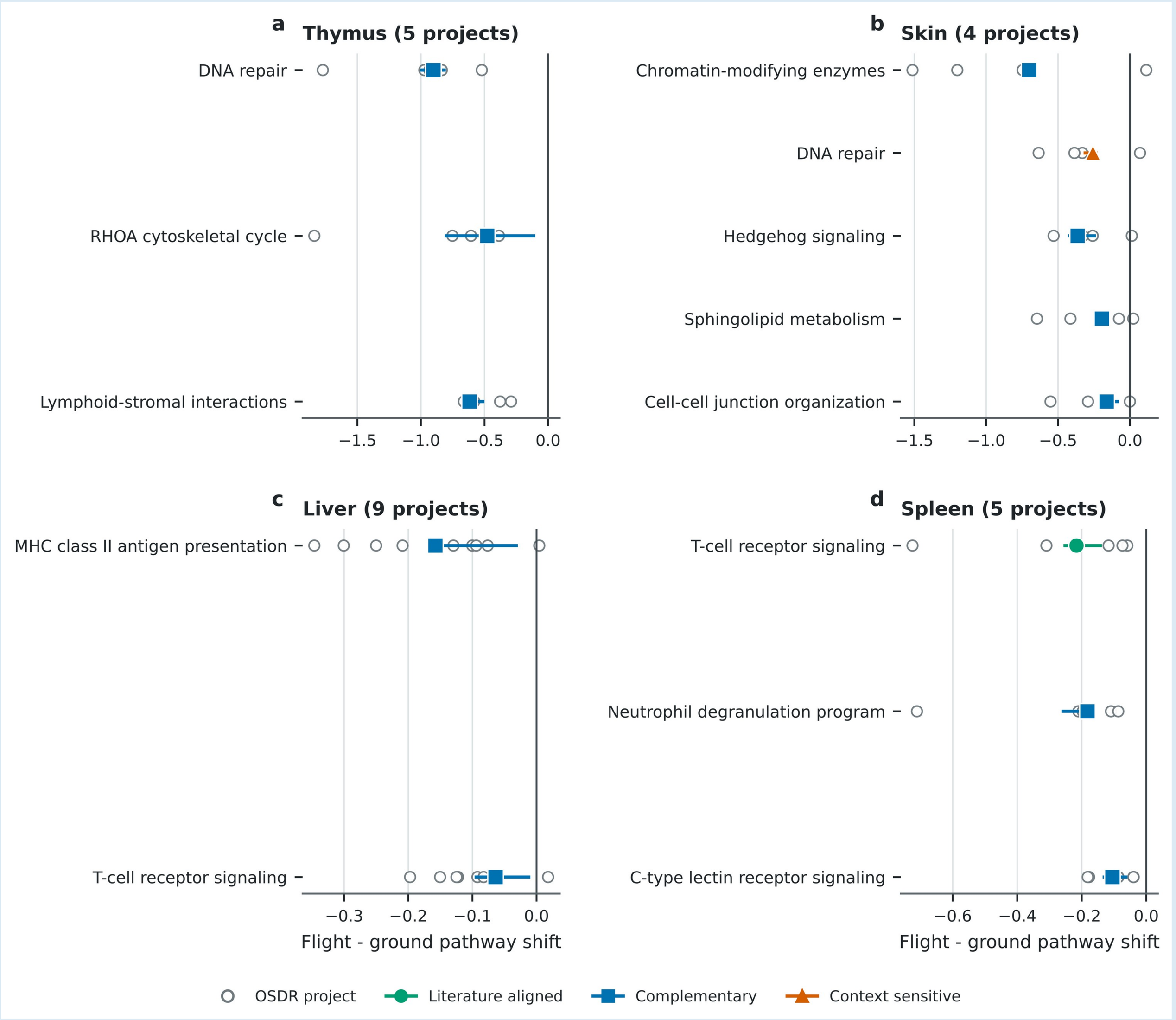
Pathway structure is wired into the decoder

Train a tissue reference, then map flight and ground samples into the same annotated program space.



approximately 2,000 HVGs | 319-387 retained Reactome programs | negative-binomial reference | 250-epoch query map

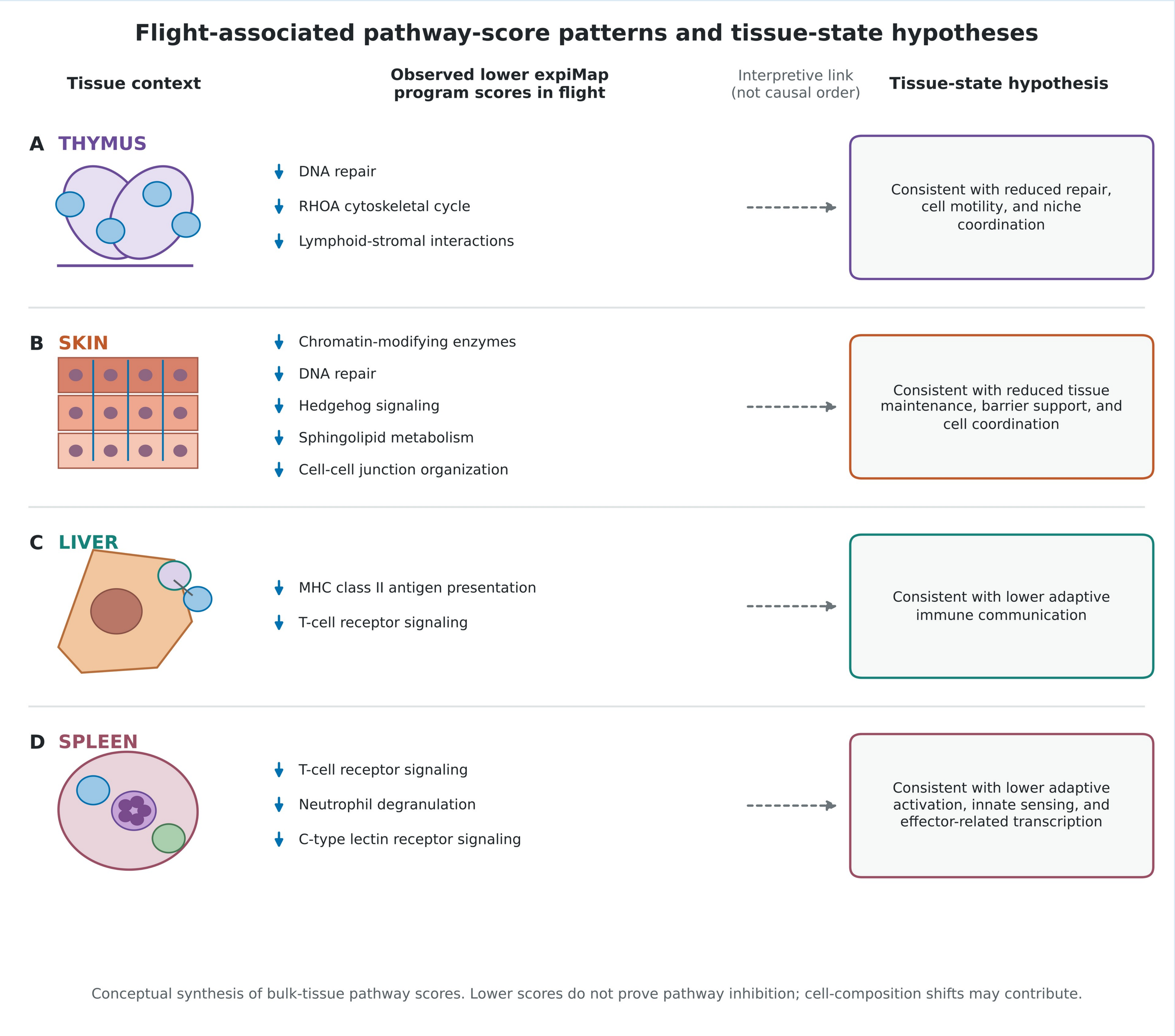
RESULTS: CROSS-MISSION PATHWAY SHIFTS



Open circles are OSDR projects; colored ranges span three complete reference-query trainings. All displayed shifts are lower in flight; axes are tissue specific.

RESULTS: BIOLOGICAL INTERPRETATION

Final robustness-filtered tissue stories



Conceptual synthesis of bulk-tissue pathway scores. Lower scores do not prove pathway inhibition; cell-composition shifts may contribute.

CONCLUSIONS

- Spleen was strongest: T-cell receptor, neutrophil degranulation, and C-type lectin receptor programs were lower across five projects and three trainings; all had GSEA FDR <0.05.
- Skin and thymus emphasized lower tissue-maintenance programs; liver added a lower adaptive-immune axis to heterogeneous metabolism.
- Bulk latent scores are testable hypotheses, not causal or cell-type-resolved pathway activity.

REFERENCES

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github.com/jasont314/nasa-mouse